

User Manual

ROBO-GOOSE

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TABLE OF CONTENTS

Contents

1. EXECUTIVE SUMMARY	2
1.1 PROJECT OVERVIEW.....	3
1.2 PURPOSE OF THIS DOCUMENT	3
2. INTENDED USAGE	3
2.1 ENVIRONMENT AND TERRAIN	3
2.2 USER QUALIFICATIONS AND REQUIREMENTS.....	3
2.3 BATTERY MONITORING, CHARGING, AND USAGE	4
3. SAFETY WARNINGS & PRECAUTIONS.....	5
Operating Restrictions	6
4. PRE-OPERATION BATTERY SAFETY CHECK.....	7
5. OPERATING INSTRUCTIONS.....	7
Startup Procedure	7
Confirming Correct Operation	7
Shutdown Procedure	8
6. CONTROLS REFERENCE	8
7. CONNECTING TO THE WEB SERVER	11
Common Issues	12
8. TROUBLESHOOTING	12
The robot won't drive at all (or only one wheel spins)	13
Controller won't pair	13
No sound from the speaker	13
The web server can't be reached.....	13
Wheels feel loose or wobble.....	14
9. MAINTENANCE, STORAGE & TRANSPORT.....	14
Storage.....	15

Transport.....	15
Outdoor Use	15

1. EXECUTIVE SUMMARY

1.1 PROJECT OVERVIEW

ROBO-GOOSE is the inaugural project for the CSE Showcase initiative. Its main purpose is to highlight the strengths of the Computer Engineering major and distinguish it from Computer Science within the University at Buffalo's CSE Department. The robot will be showcased during departmental events such as the CSE Open House and Accepted Students Day, where the Computer Engineering sub-department can use it as a conversation piece to demonstrate the hardware/software integration approach and defines and differentiates Computer Engineering from other majors.

1.2 PURPOSE OF THIS DOCUMENT

This user manual describes how to use and operate the ROBO-GOOSE. This document also describes basic debugging steps that can be used when the ROBO-GOOSE is not operating as intended. This document also describes critical safety information that all handlers of the ROBO-GOOSE must be aware of.

2. INTENDED USAGE

2.1 ENVIRONMENT AND TERRAIN

The ROBO-GOOSE is intended to be used on flat contiguous smooth surfaces such as tile floors, hardwood floors, and carpet. The ROBO-GOOSE also handles well over slightly bumpy terrain such as outside on the concrete pathways on UB North Campus, although the ROBO-GOOSE is not designed or intended to operate across such surfaces.

The ROBO-GOOSE should never be driven over, into, or near any water or liquid.

The ROBO-GOOSE is not tested to drive over dirt, grass, or any form of natural or artificial turf.

While the ROBO-GOOSE may be operated outdoors, it is not shielded from precipitation or elements. The ROBO-GOOSE should never be operated under any form of precipitation or fog, no matter how minor. The ROBO-GOOSE should not be operated outdoors when there is any chance of storm or lightning.

The ROBO-GOOSE may drive slowly across smooth ramps, such as accessibility ramps around UB campus. The ROBO-GOOSE should only be driven up or down inclines at slow or moderate speeds. The ROBO-GOOSE should never be driven across any incline of greater than fifteen-degrees.

2.2 USER QUALIFICATIONS AND REQUIREMENTS

The ROBO-GOOSE is an engineering project that was **not designed for general use. Non-technical people should not operate the ROBO-GOOSE under any circumstances.** In order to successfully operate the ROBO-GOOSE safely, the operator must have a technical engineering background. It is expected that competent engineering majors of Junior or greater standing can generally operate the ROBO-GOOSE safely that have experience with electronics after additional LiPo safety training. Extreme care and caution must always be used when operating the ROBO-GOOSE. The operator of the ROBO-GOOSE must have a solid understanding of electrical circuits, including the concepts of short circuit, power, and ground. The operator must have a thorough understanding of proper and safe usage of Lithium Polymer batteries.

2.3 BATTERY MONITORING, CHARGING, AND USAGE

The ROBO-GOOSE uses a three-cell lithium polymer battery to charge the entire robot.

Lithium Polymer (LiPo) batteries are advantageous because they provide a high energy density, can supply a large output current, are lightweight, and relatively affordable. Lithium Polymer batteries are highly volatile. LiPo batteries can spontaneously combust into chemical fires that also produce toxic chemical fumes within seconds if not used properly. Lithium Polymer batteries must constantly be monitored when in use. The ROBO-GOOSE and its battery must only be used by qualified persons who understand safe handling and usage of LiPo batteries.

This usage guide helps to understand how to use the LiPo battery on the ROBO-GOOSE but is not comprehensive. Any new handlers of the ROBO-GOOSE should first contact UB's Environment, Health & Safety (EH&S) with up-to-date guidance and training on safe handling of LiPo batteries. As of Spring 2026, EH&S can be contacted here:

<https://www.buffalo.edu/administrative-services/managing-facilities/environment-health-and-safety.html>.

This battery should only be fully charged when in use or upon imminent use. At all other times the battery should be put into storage mode. When being stored and when not in use, the battery should be stored in a safe containment unit. This Lithium Polymer battery must be stored in an electrically neutral compartment or box. The battery should not be stored alongside any other electrical components or metal components. The battery should never be placed near any sharp objects or liquids. The LiPo should never be left out on a cluttered or dirty workstation. It is recommended to store the LiPo battery in a containment box specifically designed to contain a LiPo explosion, such as a Bat-Safe box.

The Battery should only ever be charged with a purpose-built LiPo charger module (such as the iMars D300) using a balance lead. Before charging the LiPo, perform a visual inspection. If the LiPo appears to be swelling, is puffy, is scratched, dented, or injured in any other way it should

be considered damaged and not safe to use. When charging the LiPo battery, first ensure that the charger shows that all three cells are detected. Then, verify that all cells individually read above 3.2 volts *at minimum*. If any of the three cells are not detected, or if any of the cells read at below 3.2 volts, that means the battery has likely been damaged and is no longer safe to use.

Please be aware that the recommended charging conditions can vary by battery. It is required to double check the reference material for any LiPo battery being charged before charging to ensure the battery is being charged with the correct settings. The battery must be constantly supervised when charging. It is recommended, if possible, to charge the LiPo inside of a purpose-built LiPo containment box, such as Bat-Safe, which includes a dedicated port on the top to loop a charging cable into. LiPo batteries are highly volatile when being charged and need to be closely monitored. If the battery changes in size, pops, starts smelling (oftentimes described as a fruity scent), swells, or experiences any other abnormal change in behavior it must immediately stop being charged, and placed in a safe container such as Bat-Safe.

If charging the provided LiPo battery for RoboGoose v1, set the Select Task option to *Charging*. The Battery Type option should be set to *LiPo*. The LiPo battery currently used by the RoboGoose v1 should be charged to a Target Voltage of 4.2v. The Target Current determines how fast the battery would be charged and can be set to as low as 2A for a very slow charge and up to 10A for a fast charge. It is recommended to charge the battery at a rate of around 7A.

When no longer using a LiPo, it should be charged to its storage state. For optimal battery life and safety, LiPo batteries should not be left for more than a day outside of storage state. LiPos left for a long period of time outside of storage conditions can rapidly deteriorate and become a hazard or simply lose capacity. LiPo batteries set to storage conditions are also safer and less likely to spontaneously combust while being unused. To set a LiPo to storage conditions, insert the battery into the LiPo charger and set the Select Task to *Storage*. The charger will usually charge/discharge the battery to about 50% capacity at around 3.8v per cell, which is generally the recommended state to store a LiPo in. These setting may change depending on the specific battery being used, so always refer to the specific manufacturer documentation provided for a battery.

If the battery becomes damaged, however minor, it is unsafe to use. It should be marked in a clear but safe way to indicate to others that the battery has been damaged and should not be used or touched. The battery should then be safely stored and then disposed of as soon as possible. EH&S can be contacted to help with proper disposal of LiPo batteries. LiPo batteries require proper procedures in order to be safely disposed of. They should never simply be thrown in the trash.

Refer to the iMars D300 for more detailed information about how to charge LiPo batteries.

3. SAFETY WARNINGS & PRECAUTIONS

Read this entire section before operating RoboGoose. RoboGoose is a high-power robotics project, not a consumer product. Several of its hazards can cause fire, injury, or property damage.

Operating Restrictions

- Do not operate RoboGoose if impaired by drugs, alcohol, or prescription medication.
- Do not operate RoboGoose unless you are at least a Junior-level engineering student, or under the direct supervision of someone who is.
- Do not drive RoboGoose into people or walls. Collisions can dislodge electronics, break the neck, or jam the wheels.
- Do not pick RoboGoose up by the neck. The neck is the most fragile component on the robot and has snapped off the chassis multiple times under load. Always lift from the chassis.
- Do not exceed a slow walking pace near children, pets, or in crowds.
- Do not operate RoboGoose around real geese. Beyond the obvious, RoboGoose is not waterproof and a hostile flock will not be deterred by it.

RoboGoose v1 has unseated wires on several Arduino Mega GPIO pins that are not soldered or secured to a protoboard. These bare conductors can contact each other, adjacent metal chassis parts, or the LiPo battery terminals. A short circuit on the 12 V rail can damage the Spark MAX controllers, the buck converter, or the Mega, and may cause rapid uncontrolled battery discharge. Do not operate the robot with loose wiring that is not fully isolated and strain relieved. **Before every session, visually inspect every GPIO connection. Do not operate if any conductor is exposed or unsecured.**

Wiring routed along the exterior of the chassis is accessible to bystanders, including children who may be present at showcase events. The open rear of the shell exposes live conductors. Only operate RoboGoose in controlled demonstration areas where an authorized operator maintains line-of-sight at all times. Do not leave the robot powered and unattended near the public.

The 3S LiPo battery can spontaneously combust into a chemical fire producing toxic fumes if punctured, over-discharged, short-circuited, or otherwise damaged. See § 2.3 and § 4 for full handling procedures. **Never operate RoboGoose without first completing the pre-operation battery safety check.**

4. PRE-OPERATION BATTERY SAFETY CHECK

Before turning on RoboGoose, ensure that the battery is appropriately charged and there are no loose wires. Before connecting the battery to the LiPo connector, check for any visible scratches. If you encounter any scratches, remove the battery entirely and place it in a safe place. Do not operate RoboGoose if anything appears to be damaged. Contact the current maintainers of RoboGoose.

If connecting the battery results in any smoke, disconnect the battery, place it in a safe place, and contact UBPD.

5. OPERATING INSTRUCTIONS

Startup Procedure

1. **Inspect the robot.** Confirm all GPIO wires are seated, no exposed conductors are touching metal, and the neck and shell are secure. Wheels should spin freely when rotated by hand.
2. **Complete the pre-operation battery safety check** (see § 4). Do not skip this step.
3. **Connect the LiPo battery** to the main power connector inside the chassis. Confirm the connector is fully seated.
4. **Wait for boot.** Within a few seconds, you should observe:
 - a. The startup sound plays from the speaker (a random track from folder 01 on the SD card).
 - b. The RGB LED eyes turn on and begin their breathing animation.
 - c. The LCD on the chassis lights up and displays motor RPM (Mode 0 by default).
 - d. The Bluetooth Nano ESP32 begins searching for an Xbox controller.
5. **Pair the controller.** Hold the **sync button** on the back of the Xbox controller (next to the charging port) until the Xbox logo on the front flashes rapidly. After a few seconds, the Nano will pick it up and the controller's light will go solid. RoboGoose is now drivable.
***Note:** The Xbox button on the front of the controller powers the controller on. It does not pair the controller. Pairing requires the sync button on the back.*

Confirming Correct Operation

After startup, perform a brief functional check before demonstrating:

- Push the left stick forward gently: both wheels should spin smoothly.
- Push the right stick left or right: the robot should turn smoothly.
- Press A, B, X, Y in turn: eyes should change color and sound effects should play.

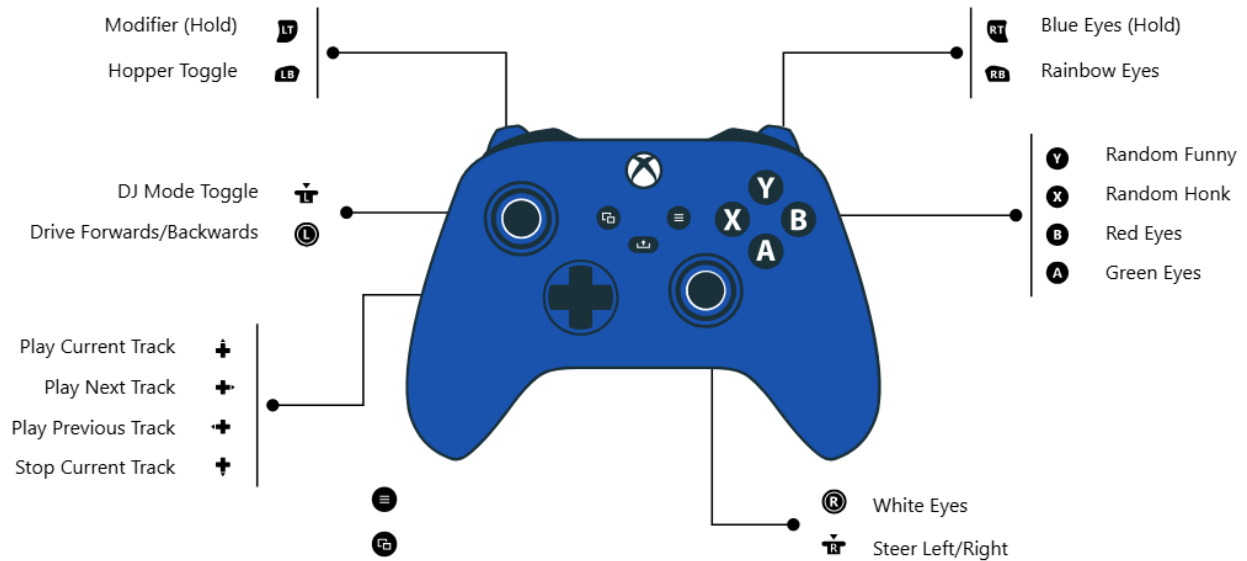
If any of these fail, see § 8 (Troubleshooting).

Shutdown Procedure

1. Bring RoboGoose to a complete stop on a level surface.
2. Disconnect the LiPo battery from the main power connector. **The battery should be disconnected after every use**
3. If storing for more than 24 hours, place the battery on its charger and run a Storage cycle (see § 2.3).
4. Store RoboGoose in a cool, dry place.

6. CONTROLS REFERENCE

The Xbox controller pairs via Bluetooth to the Nano ESP32, which sends inputs to the Mega over Serial1. If the controller is not paired, hold the sync button (next to the charging port) to pair it to the Nano ESP32. LT acts as a modifier: when held, most buttons do something different.



LT + B	Pan Noise
LT + A	Apple Pay Noise
LT + +	Volume Up
LT + -	Volume Down
LT + ⏮	Previous LCD Mode
LT + ⏭	Next LCD Mode

Controls without LT held:

Input	Action
Left stick [up/down]	Drive forward/back
Right stick [left/right]	Steering
A	Green eyes
B	Red eyes
X	Random honk
Y	Random sound
LB	Toggle hopper servo
RB	Rainbow eyes
LS click	Toggle DJ mode
RS click	White eyes
D-pad Left/Right	Previous/next music track
D-pad Up	Replay current track
D-pad Down	Stop audio
RT (held)	Blue eyes

Controls with LT held (modifier):

Input	Action
LT + A	Play annoying sound 03/001 (Apple Pay)
LT + B	Play annoying sound 03/004 (Pan)
LT + D-pad Up	Volume up
LT + D-pad Down	Volume down
LT + D-pad Left	Previous LCD screen mode
LT + D-pad Right	Next LCD screen mode

The LCD Screen on the robot has five display modes, cycled by holding LT and pressing D-pad left or right:

Mode	Display
0	Left and right motor RPM
1	Left and right motor temperatures
2	Battery voltage, battery percentage, current volume
3	Current music track, DJ mode state, current volume
4	Live IMU pitch and forward tilt (switch off when driving)

About Mode 4: This mode refreshes 3x faster than the others so tuners can watch the pitch value change in real time. The faster refresh makes driving noticeably less responsive. **Switch off Mode 4 before driving the robot in a demo.** It is not unsafe to do so, just a lot less smooth.

7. CONNECTING TO THE WEB SERVER

To connect to the built-in web server, first ensure the robot is on and that the server-dedicated Arduino Nano ESP32 (The one with the black case, NOT BLUE) is receiving power. Once powered, it will automatically begin hosting the server within milliseconds.

Network Details:

- Network Name: **Goosebot**
- Password: **pizzamov!e** (as of writing this document)
- URL: **goosebot.io**

Next, open the network/WiFi settings on your preferred device. Within the networks list, you should see “GooseBot”, select this option. Once selected, you will be prompted for a password. This password can be changed via a string in the Nano’s code (refer to the design documentation or the code directly). The current password is ‘pizzamov!e’. Once entered, your device should connect to the server.

After connecting, some devices will automatically open the web app page (goosebot.io). However, if your device does not, simply open your browser and enter the URL manually. As long as you are on the network, all functions on the website should work as intended. Here you may manage the current song, volume, or play a sound via the soundboard. The app also displays current telemetry (like battery, motor temps, rpms, and IMU readings) in a dedicated page.

Common Issues

- **"My phone says there's no internet."** The GooseBot network is fully self-contained and has no uplink to the outside internet. Your phone is working normally. The soundboard will function regardless.
- **Captive portal didn't pop up.** Type **goosebot.io** manually in your browser. Some operating systems (notably newer iOS) suppress captive portals on networks they consider "unrecognized."
- **Page won't load.** Confirm you are still connected to the GooseBot network. Phones often auto-switch back to a saved cellular or home network when no internet is detected.

8. TROUBLESHOOTING

If experiencing any operational failure, **disconnect the LiPo battery first** if you observe smoke, smell anything fruity or chemical, or see the battery swell. Place the battery in a safe containment box and contact a qualified maintainer.

For non-emergency issues, the table below covers the most common operator problems. For deeper hardware-level debugging, refer to the Final Design Document § 8.

The robot won't drive at all (or only one wheel spins)

- Confirm RoboGoose is not in death mode (see § 6). Pick it up and set it upright on a level surface, and wait a few seconds for motors to resume.
- Confirm the battery is connected and charged. If LCD Mode 2 shows below ~ 10.2 V, the battery should be swapped.
- Press the reset button on the Mega. A reset can clear any locks on the Mega.
- If only one wheel spins, the issue is most likely a loose CAN wire or a Wago connector that slipped. Power down and reseal all CAN and motor connections.

Controller won't pair

- Confirm the Xbox controller has charged batteries. Replace the AA batteries if in doubt, the project uses Panasonic rechargeables that lose pairing range when low.
- Hold the **sync button** on the back of the controller (next to the charging port) for several seconds until the Xbox logo flashes rapidly.
- Power-cycle RoboGoose by disconnecting and reconnecting the LiPo battery, then retry pairing.
- Confirm the Bluetooth Nano ESP32 (the blue-cased one) is receiving power. A loose 5V or GND connection will silently disable controller input.

No sound from the speaker

- Confirm the Micro SD card is seated in the DFPlayer Mini. If it has been removed, the speaker will be silent with no error indication. Reseat the Micro SD Card.
- Confirm volume is not muted. Use LT + D-pad Up to raise volume. Switch to LCD Mode 2 or 3 to see the current volume value.
- Confirm the speaker is connected to the amplifier output terminals.

The web server can't be reached

See § 7 for common issues. Most failures are network-side on the user's device, not on RoboGoose. Turn WiFi off and on on your device and reset the Mega if it keeps happening.

Wheels feel loose or wobble

Inspect the leg to chassis screws and the wheel to gearbox screws. PLA is brittle under repeated load and threaded holes can strip over time. Reseat or replace screws as needed.

If regular operation is not restored after the steps above, disconnect the LiPo battery. **Do not attempt to re-configure the wiring, disconnect fuses, or tamper with electronics.** Store RoboGoose in a cool, dry place, and contact the maintainers or Kris Schindler (kds@buffalo.edu) for further assistance.

9. MAINTENANCE, STORAGE & TRANSPORT

Storage

- Disconnect the LiPo battery and store it in a Bat-Safe or equivalent fireproof containment box. See § 2.3 for full battery storage guidance.
- Store RoboGoose itself in a cool, dry, indoor location such as the Baldy 200C Lab. The shell is PLA and will warp if exposed to high heat (cars in summer, near radiators, etc.).
- If storing for more than a week, also disconnect the battery from any chargers and confirm it is at storage voltage (~3.8 V/cell).

Transport

- **Always disconnect the LiPo battery before transporting RoboGoose.**
- Transport the battery separately in its containment box. Do not transport a charged LiPo in a closed car on a hot day.
- Lift RoboGoose by the chassis, never by the neck or shell.
- If transporting in a vehicle, secure RoboGoose so it cannot roll. The wheels have no parking brake.

Outdoor Use

RoboGoose is not designed to function outdoors, though it can be operated outdoors on smooth surfaces such as concrete or asphalt sidewalks. Note that RoboGoose has not been tested on rough terrain, grass, or non-level ground. Do not operate on such surfaces.

Do not operate outdoors during precipitation, fog, or any chance of storms.

For information regarding how to handle the battery when RoboGoose is not in use, refer to § 2.3 BATTERY MONITORING, CHARGING, AND USAGE.